

WEEK 4 – POWER BI STOCK ANALYSIS

1. INTRODUCTION

This week, the Shopify stock market dataset was further analyzed using Microsoft Power BI. The main focus was on calculating **20-Day and 50-Day Simple Moving Averages (SMA)** and creating a **Moving Average Crossover Signal** using DAX. Visualizations, timeline slicers, dynamic switches, conditional formatting, and a custom tooltip page were also added to make the dashboard interactive and informative.

2. DAX MEASURES

DAX measures were created to calculate the moving averages and identify stock trends.

20-Day Simple Moving Average

The 20-Day SMA was calculated using the average trading volume over the selected 20-day period.

Volume 20-Day Simple Moving Average =

VAR CurrentDate = MAX(Dim_Date[Date])

VAR PeriodDays = 20

VAR DateInPeriod =

```
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )
```

RETURN

```
CALCULATE(  
    AVERAGE(Shopify_Stock[Volume]),  
    DateInPeriod  
)
```

50-Day Simple Moving Average

The 50-Day SMA was calculated in the same way using a 50-day period.

Volume 50-Day Simple Moving Average =

VAR CurrentDate = MAX(Dim_Date[Date])

VAR PeriodDays = 50

VAR DateInPeriod =

```
DATESINPERIOD(  
    Dim_Date[Date],  
    CurrentDate,  
    -PeriodDays,  
    DAY  
)
```

RETURN

```
CALCULATE(  
    AVERAGE(Shopify_Stock[Volume]),  
    DateInPeriod  
)
```

Moving Average Crossover Signal

A crossover signal was created to identify whether the stock trend was **Bullish** or **Bearish**.

MA Crossover Signal =

```
IF(  
  ISBLANK([20 SMA]) ||  
  ISBLANK([50 SMA]),  
  "No Signal",  
  IF(  
    [20 SMA] > [50 SMA],  
    "Bullish (20 > 50)",  
    "Bearish (20 < 50)"  
  )  
)
```

3. STOCK PRICE AND VOLUME VISUALIZATION

A **Line Chart** was created to display the stock price and moving-average trends over time.

- **X-Axis:** Date
- **Y-Axis:** Close Price, 20 SMA and 50 SMA
- **Legend:** Different measures were used to compare the trends.
- **Zoom Slider:** Added to allow users to focus on a specific time period.

This visualization helps to understand changes in Shopify's stock price and identify the relationship between the 20-day and 50-day moving averages.

4. TRADING VOLUME AND VOLATILITY AREA VISUAL

A **Clustered Column Chart** was created to analyze trading volume and volatility.

- **X-Axis:** Date
- **Y-Axis:** Total Volume and Volume 20 SMA
- Conditional formatting was applied to highlight volume spikes.

A conditional formatting measure was also created to identify unusually high trading volume. This makes significant volume changes easier to identify in the chart.

5. TIMELINE SLICERS AND DYNAMIC SWITCHES

Interactive controls were added to make the dashboard easier to explore.

Date Range Slicer

A **Date Range Slicer** was added using the Dim_Date[Date] field in **Between** mode. This allows users to select a specific date range.

Quick Period Buttons

Quick period buttons were added using field parameters to provide options such as:

- 1 Month
- 3 Months
- 6 Months
- 1 Year
- Year-to-Date (YTD)
- 1 Year

Parameter Switch

A parameter switch was also created to allow users to select different stock price measures dynamically.

6. CUSTOM TOOLTIP PAGE

A custom tooltip page named **Stock_Tooltip** was created to provide additional information when the user hovers over a data point.

The tooltip displays selected stock details such as:

- Open Price
- High Price
- Low Price
- Close Price
- Volume
- Moving Average information

This improves the interactivity of the dashboard and allows users to view detailed information without leaving the main report page.

7. CONCLUSION

Overall, Week 4 focused on improving the Shopify Stock Analysis dashboard using **DAX measures and interactive Power BI features**. The 20-Day and 50-Day moving averages helped identify stock trends, while the crossover signal provided a simple indication of bullish and bearish movements. Timeline slicers, dynamic switches, conditional formatting, and the custom tooltip made the dashboard more interactive and user-friendly.

